Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *EU Pharmaceutical Regulatory NLP — French*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

DrBenchmark's task taxonomy (POS, NER, MCQA, MCC, MLC, STS) is calibrated for clinical and research NLP rather than EU pharmaceutical regulatory workflow categories [Q81, Q83]. The deployment requires NER and STS over EU regulatory document genres (SmPCs, PILs, CTD modules, EU-RMPs), but the benchmark organizes tasks around general biomedical constructs and includes substantial irrelevant categories: MCQA over pharmacy exam questions [Q39], abstract-level specialty classification [Q38], ICD-10 chapter classification [Q45], and negation/speculation scope [CAS, ESSAI]. The dataset analysis confirms critical task-type mismatches for FrenchMedMCQA, MORFITT, and DiaMED, which evaluate constructs orthogonal to the deployment's NER/STS extraction needs (FRENCHMEDMCQA-D11, MORFITT-D7, DIAMED-D6, DIAMED-D13). No category covers regulatory equivalence scoring as a distinct output decision rule, despite this being a HIGH-priority deployment dimension. Partial overlap exists for NER (QUAERO, MANTRAGSC, DEFT2021, PxCorpus) and STS (CLISTER, DEFT2020), so the benchmark is not entirely orthogonal — but construct underrepresentation for regulatory-specific task types is severe.

ID	Evidence
Q38	"MorFITT (Labrak et al., 2023) is a multi-label dataset annotated with medical specialties. It contains 3,624 biomedical abstracts from PMC Open Access. It has been annotated across 12 medical specialties (more detail in Appendix B.5), for a total of 5,116 annotations." (p.4)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
Q83	"In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain." (p.9)
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
MORFITT-D7	La mise en place par cathétérisme d'une valve aortique (TAVI)… Le registre France-TAVI… va inclure plus de 3 000 cas — Abstract-level classification only; no token-level entity annotations for NER task
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: — MCQA format — tests knowledge retrieval, not NER or STS; task type mismatch
DIAMED-D13	Il s'agit d'un adolescent âgé de 16 ans, sportif (tennis) qui se plaignait depuis plusieurs mois de douleurs mécaniques du coude droit. — Clinical case narrative about orthopedic condition; no pharmaceutical content

Strengths

- Task inventory includes both NER and STS, the two task types the deployment requires [Q1, Q4]

- MANTRAGSC fr_emea and fr_patents subsets provide task categories that overlap genre-wise with the deployment's regulatory and IP workflow components (MANTRAGSC-D1, MANTRAGSC-D7)
- DEFT2020 STS Task 1 contains drug leaflet sentence pairs as a sub-component, structurally analogous to safety-warning equivalence checking (DEFT2020-D1, DEFT2020-D2)

ID	Evidence
Q1	“It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification.” (p.1)
Q4	“These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS).” (p.1)
MANTRAGSC-D1	Chaque comprimé contient 500 mg de ranolazine. — Drug composition statement with INN annotated CHEM; EMEA register
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
DEFT2020-D2	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — Drug safety warning with dose-time threshold — deployment-central entity type
MANTRAGSC-D7	Forme posologique orale selon la revendication 1, dans laquelle ledit antagoniste opioïde libérable est la naltrexone et ledit opioïde libérable est la codéine, dans laquelle le rapport de la naltrexone sur la codéine libérable est de 0,005 : 1 à 0,044 : 1. — French patent claim with INNs and dosage ratio; patent claim register confirmed

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required test case categories for the deployment: (a) NER over EU regulatory document genres (SmPCs, PILs, CTD modules, EU-RMPs) for INN/ATC/excipient/posology/contraindication entity types; (b) regulatory-equivalence STS over EMA-templated text; (c) potentially patent-claim NER for IP workflow component. Per elicitation A1–A3 these are the central use case.

IO-2: Multiple regionally-relevant categories are omitted: (i) regulatory-equivalence STS as a distinct decision rule is absent — STS in the benchmark targets general semantic proximity [Q41]; (ii) no task category covers SmPC, CTD module, or EU-RMP NER as a genre — DrBenchmark acknowledges no large French biomedical benchmark existed prior [Q16] and web search confirms no French NLP benchmark with confirmed CTD module text exists [WEB-6]; (iii) IDMP-driven structured extraction from regulatory text is unaddressed [WEB-16, WEB-17].

IO-3: Several included categories are largely irrelevant to the deployment: pharmacy-diploma MCQA evaluates declarative knowledge retrieval rather than NER/STS (FRENCHMEDMCQA-D11, FRENCHMEDMCQA-D14); abstract-level multi-label specialty classification (MORFITT, DEFT2021-cls) does not test span-level extraction (MORFITT-D7, DEFT2021-D16); ICD-10 chapter classification (DiaMED) is orthogonal to regulatory entity extraction (DIAMED-D6, DIAMED-D13); negation/speculation scope NER (CAS, ESSAI) is at most indirectly relevant (CAS-D5, ESSAI-D12).

IO-4: Category gaps harming content validity: absence of regulatory-equivalence STS task category; absence of SmPC/CTD/EU-RMP genre-specific NER task category; absence of IDMP-structured extraction task. Web search confirms these are field-level gaps, not just benchmark gaps [WEB-11, WEB-16].

ID	Evidence

Q1	“It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification.” (p.1)
Q4	“These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS).” (p.1)
Q16	“To our knowledge, aside the multilingual benchmark BigBIO (Fries et al., 2022) which includes only 4 corpora for French and is initially intended for generative text completion under zero-shot scenario, no large benchmark specialized in the French biomedical field exists.” (p.2)
Q39	“FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question.” (p.4)
Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
Q81	“Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks.” (p.8)
CAS-D5	tokens: ['sans', 'antécédent', 'familial', 'de', 'maladie', 'colique']; ner_tags: [0, 1, 2, 2, 2, 2] — Negation scope spanning clinical history phrase; shows annotation granularity
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
MORFITT-D7	La mise en place par cathétérisme d'une valve aortique (TAVI)… Le registre France-TAVI… va inclure plus de 3 000 cas — Abstract-level classification only; no token-level entity annotations for NER task
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: — MCQA format — tests knowledge retrieval, not NER or STS; task type mismatch
ESSAI-D12	le traitement par un inhibiteur de PARP, le talazoparib, peut être plus efficace et mieux toléré de la chimiothérapie de référence — Speculation NER marks hedging scope, not INN entity
DIAMED-D13	Il s'agit d'un adolescent âgé de 16 ans, sportif (tennis) qui se plaignait depuis plusieurs mois de douleurs mécaniques du coude droit. — Clinical case narrative about orthopedic condition; no pharmaceutical content
FRENCHMED MCQA-D14	Parmi les propriétés suivantes du monoxyde de carbone, indiquer celle(s) qui est (sont) exacte(s) — Classification of true/false propositions — no STS pairwise similarity signal
DEFT2021-D16	features: id, document_id, text, specialities, specialities_one_hot — No STS pair structure; classification only; cannot evaluate regulatory equivalence scoring
WEB-6	DrBenchmark paper confirms task inventory of 20 tasks across the five categories; no SmPC/CTD/EU-RMP-specific task category present (https://arxiv.org/html/2402.13432v1)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)
WEB-17	https://www.linguamatics.com/blog/countdown-idmp-compliance-are-you-ready

Information Gaps

- No documented evaluation category for regulatory-equivalence STS or IDMP-structured extraction; whether the benchmark's general STS performance is at all predictive of regulatory-equivalence performance is unknown.

Requires Expert Verification

- Whether deployment teams treat MANTRAGSC fr_patents as adequate proxy for pharmaceutical patent claim NER given its 20-example test set.

Remediation

Gap 1: No task category covers IDMP-driven structured extraction or regulatory-equivalence STS, both of which are central to the deployment [WEB-11, WEB-16].

Recommendation: Add two task categories to the evaluation suite: (a) IDMP attribute extraction from SmPC text targeting ATC, INN, excipient dose, therapeutic indication; (b) regulatory-equivalence STS over EMA-templated safety-warning pairs. Both can be designed as extensions of the existing HuggingFace toolkit [Q48].

ID	Evidence
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)